

AP Environmental Science: The Complete Exam Guide

All 9 units explained, the three free-response question types, a step-by-step math playbook, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**2:40**

hours total
90 + 70 minutes

80+3

multiple-choice +
free-response questions

60%

of your score is
multiple choice

69%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) is weighted towards the multiple-choice section (60%), with three free-response questions making up the other 40%. A calculator is allowed throughout.

60%

Section I • Multiple choice • 80 questions

40%

Section II • Free response • 3 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	60% of score 1 hr 30 min 80 questions	Individual questions and sets built on tables, charts, graphs, models, maps and short texts. Explain concepts, analyze data and propose solutions.
II • Free response	40% of score 1 hr 10 min 3 questions	Q1 Design an investigation • Q2 Analyze an environmental problem and propose a solution • Q3 Analyze a problem and propose a solution using calculations.

FULLY DIGITAL

Everything happens in Bluebook

You answer every question, including the written responses, in College Board's **Bluebook** testing app. Responses are submitted automatically at the end. Practice typing timed answers before May.

CALCULATOR ALLOWED

But show your math

A calculator is permitted on the whole exam. On calculation questions you must still **show your setup and units** to earn the points.

How to use this guide

Just starting?

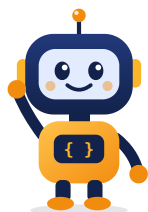
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 15 minutes, then use the final-four-weeks plan.



Pigo's tip: APES is about **connections**: how one change ripples through an ecosystem or a human system. Practice "cause → effect → consequence" chains.

SECTION I

02 The multiple-choice section

80 questions in 90 minutes gives you about 1 minute 7 seconds per question. Speed matters more here than on most AP science exams.

80

questions, many in data-based sets

~1:07

minutes per question on average

60%

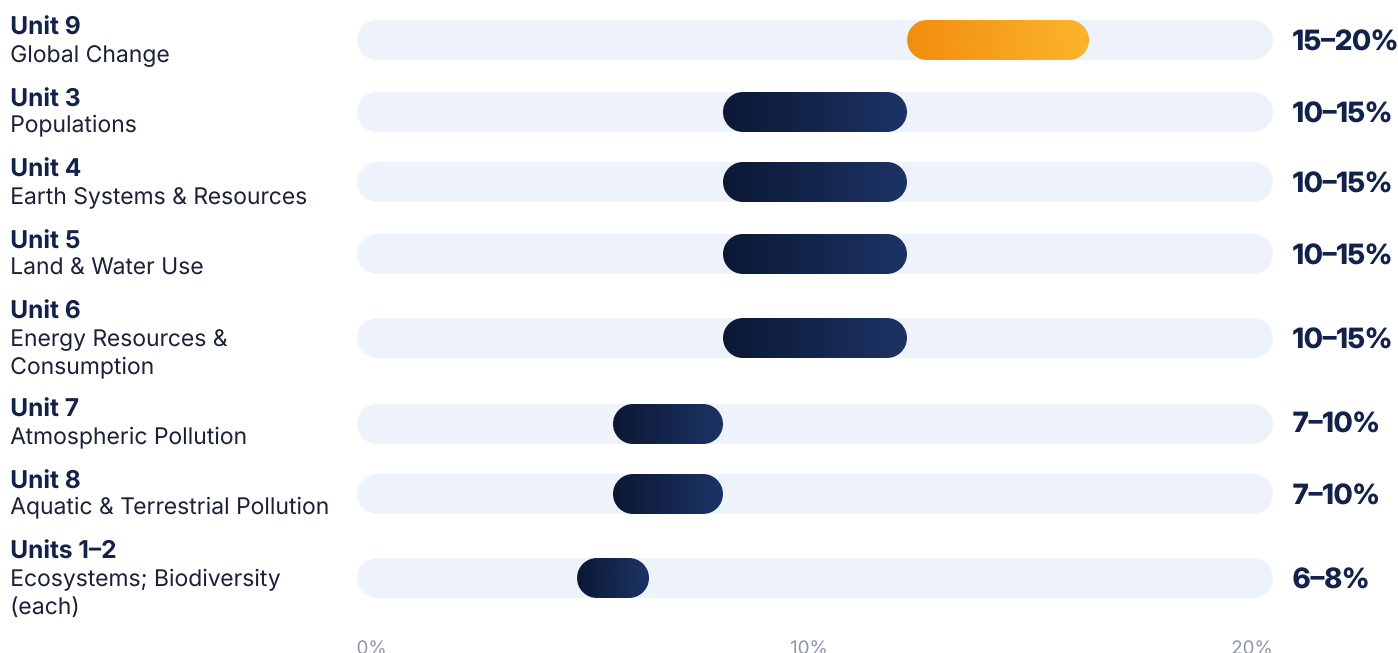
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.



What this means for your study time: Global change (climate, ozone, invasive species) is the single biggest unit, and Units 3–6 together are about half of the section.

Strategies that work on this section

- **Keep moving.** At about 1 minute per question, don't let one data set eat five minutes.
- **Read the axes and units first** on every graph or table.
- **Know the classic examples:** eutrophication, ozone depletion, acid rain, the tragedy of the commons.
- **Choose the most specific answer** that is directly supported by the data shown.
- **Watch for "most likely" and "best":** several choices may be true, but only one is best.
- **Never leave a blank.** Guess and flag, then return if time allows.

03 WHAT THE COURSE COVERS

The 9 units, in plain English

Everything on the exam fits into these nine units.

UNIT
1
6–8%

The Living World: Ecosystems

Be ready to: describe biomes; trace carbon, nitrogen, phosphorus and water cycles; calculate energy flow with the 10% rule.

UNIT
2
6–8%

The Living World: Biodiversity

Be ready to: explain genetic, species and ecosystem diversity; ecosystem services; island biogeography; ecological succession.

UNIT
3
10–15%

Populations

Be ready to: compare r- and K-selected species; read survivorship curves and age-structure diagrams; calculate growth rate and doubling time; explain demographic transition.

UNIT
4
10–15%

Earth Systems & Resources

Be ready to: explain plate tectonics, soil formation, atmosphere layers, global wind patterns, watersheds and El Niño.

UNIT
5
10–15%

Land & Water Use

Be ready to: compare agricultural practices, irrigation, pest control, forestry, mining and urbanization; explain the tragedy of the commons and sustainable methods.

UNIT
6
10–15%

Energy Resources & Consumption

Be ready to: compare fossil fuels, nuclear and renewables; explain how electricity is generated; calculate energy use and cost; describe efficiency and conservation.

UNIT
7
7–10%

Atmospheric Pollution

Be ready to: identify primary and secondary pollutants; explain photochemical smog, thermal inversions, acid deposition and indoor pollution; describe Clean Air Act controls.

UNIT
8
7–10%

Aquatic & Terrestrial Pollution

Be ready to: explain eutrophication, dissolved oxygen, sewage treatment, bioaccumulation and biomagnification, LD50 and solid waste.

UNIT
9
15–20%

Global Change

Be ready to: explain stratospheric ozone depletion, the greenhouse effect and climate change impacts, ocean acidification, invasive species and endangered species.

Laws and treaties appear often. Know the purpose of the Montreal Protocol, Clean Air Act, Clean Water Act, Endangered Species Act and CITES.

SECTION II

04 The 3 free-response questions

70 minutes for 3 questions, each worth 10 points. The three question types are the same every year.

Q1 • 10 pts

Q2 • 10 pts

Q3 • 10 pts

30 raw points in total, worth 40% of your AP score

Q1

Design an investigation

10 points • with a model or visual

Identify a research question, hypothesis, variables and controls, then describe data collection and explain results for an environmental scenario.

Name the independent variable, dependent variable and one specific control.

Q2

Analyze a problem + propose a solution

10 points • with a model or visual

Describe environmental concepts shown in a diagram or data, explain effects, and propose and justify a solution.

A solution needs an action **and** how it reduces the problem.

Q3

Analyze a problem with calculations

10 points • math required

Use calculations (percent change, unit conversion, energy, population growth) to analyze a scenario, then propose and justify a solution.

Show setup and units at every step; the answer alone may earn nothing.

• A time plan for the 70 minutes

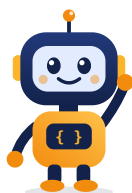
Skim
3 min

Q1
~22 min

Q2
~22 min

Q3
~23 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



When a question asks you to **"identify"**, one word can do it. **"Explain"** needs a full cause-and-effect sentence.

05 FREE-RESPONSE PLAYBOOK

Anatomy of a full-credit answer

Q3 always needs math with setup shown. Here is a short practice prompt with a full-credit answer.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Q3 style): A household runs a 1,500-watt space heater for 6 hours a day for a year. Electricity costs \$0.12 per kWh. (a) Calculate the electricity used in one year, in kWh. (b) Calculate the annual cost. (c) Propose one way to reduce this energy use and explain how it helps.

Full-credit response

① (a) $1,500 \text{ W} = 1.5 \text{ kW}$. $1.5 \text{ kW} \times 6 \text{ h/day} \times 365 \text{ days} = 3,285 \text{ kWh}$.

② (b) $3,285 \text{ kWh} \times \$0.12/\text{kWh} = \$394.20$ per year.

③ (c) Add insulation and weather-stripping around windows and doors.

④ Less heat escapes the room, so the heater runs fewer hours to keep the same temperature, reducing the kWh used and the coal or gas burned at the power plant.

- ① **Convert units first** and show the conversion.
- ② **Write every factor with units** so units cancel visibly.
- ③ **Propose a specific action**, not "use less energy".
- ④ **Explain the mechanism** linking the action to less energy use.

◆ Five habits for every free-response question

1

Match the task verb: identify, describe, explain, justify.

2

Show setup and units for every calculation.

3

Give one specific example per point; extras aren't scored.

4

Propose solutions with an action and a reason.

5

Stuck? Answer the other lettered parts; each scores separately.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

The ozone hole causes **global warming!** (It doesn't.)



01 Ozone vs climate change

Stratospheric ozone depletion increases UV-B reaching Earth. It is not the main cause of global warming.

02 Good vs bad ozone

Stratospheric ozone protects us; ground-level ozone is a secondary pollutant in photochemical smog.

03 Eutrophication sequence

Nutrients → algal bloom → algae die → decomposers use O_2 → hypoxia → fish die-off. Don't skip the decomposers.

04 Bioaccumulation vs biomagnification

Bioaccumulation builds up in one organism over time; biomagnification increases up a food chain.

05 Vague solutions

"Reduce pollution" earns nothing. Name a specific action and explain how it works.

06 Doubling time

Rule of 70: doubling time $\approx 70 \div$ growth rate in percent. Use 2, not 0.02.

07 Renewable vs clean

Biomass and hydroelectric are renewable but still have environmental impacts. Nuclear is low-carbon but not renewable.

08 Correlation as cause

Data showing two trends together doesn't prove one causes the other. Say "is associated with".

09 Missing units

A calculation answer without units or a setup may score zero on Q3.

10 Too many examples

If asked for one example, only the first one is scored. Make it your best.

Quick self-check: a country has a birth rate of 30 per 1,000 and a death rate of 10 per 1,000. What is its growth rate and doubling time? (2% per year; $70 \div 2 = 35$ years.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and practice APES math every week.



Sep – Oct

Living world



Units 1–3

Ecosystems, cycles, biodiversity and population dynamics. Learn the rule of 70 and percent change early.

Nov – Dec

Earth + land



Units 4–5

Earth systems, soil, agriculture, water use and urbanization. Start one free-response question per week.

Jan – Feb

Energy



Unit 6

Fossil fuels, nuclear and renewables. Practice energy and cost calculations (Q3 style).

Mar – Apr

Pollution + change



Units 7–9

biggest unit

Air, water and land pollution, then climate change and biodiversity loss. Begin timed 80-question sets.

Early May

Sharpen



Free-response drills

All three question types under timing. Review laws, treaties and the traps list.

Exam day

May



AP Environmental Science exam

Sleep well, charge your device, pack a calculator, and use the checklist at the end.

◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 40 multiple-choice questions in 45 minutes plus one free-response question	List your three weakest units
3 weeks out	Drill weak units; one Q3 calculation question daily	Show setup and units every time
2 weeks out	Full practice exam under timing	Finish 80 questions in 90 minutes
Final week	Light review: laws, cycles, pollutants, traps list	Rest the night before

08 TEST YOURSELF

10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **15 minutes** for all ten, then check page 10.

1 UNIT 3 • POPULATION GROWTH

A country has a crude birth rate of 30 per 1,000 and a crude death rate of 10 per 1,000, with no migration. What is its approximate doubling time?

- A 20 years
- B 35 years
- C 50 years
- D 70 years

2 UNIT 1 • ENERGY FLOW

Producers in a food chain store 10,000 kcal. About how much energy reaches the **secondary** consumers?

- A 1,000 kcal
- B 100 kcal
- C 10 kcal
- D 1 kcal

3 UNIT 8 • RADIOACTIVE DECAY

A 500 g sample of a radioactive isotope has a half-life of 10 years. How much remains after 30 years?

- A 250 g
- B 166.7 g
- C 125 g
- D 62.5 g

4 UNIT 8 • EUTROPHICATION

Fertilizer runoff enters a lake. Which sequence best describes what happens next?

- A Algal bloom → algae die → decomposers consume O_2 → fish die-off
- B Fish die-off → algal bloom → O_2 rises
- C Decomposers increase → O_2 rises → algal bloom
- D Algal bloom → O_2 rises at night → fish population grows

5 UNIT 9 • OZONE

Which statement about stratospheric ozone depletion is correct?

- A It is caused mainly by CO_2 and is the main driver of warming
- B It increases ground-level smog
- C It was caused by sulfur dioxide from coal plants
- D Chlorine from CFCs breaks down ozone, increasing UV-B reaching the surface

6 UNIT 7 • SMOG

Photochemical smog in a city usually peaks in the early afternoon. What best explains this timing?

- A Nighttime cooling traps pollutants
- B Rain washes pollutants out in the morning
- C Morning traffic releases NO_x and VOCs, and sunlight drives reactions that form ozone
- D Power plants only operate in the afternoon

7 UNIT 6 • ENERGY MATH

A 1,500 W heater runs 6 hours a day for 365 days at \$0.12 per kWh. What is the annual cost?

- A \$65.70
- B \$197.10
- C \$394.20
- D \$1,080.00

8 UNIT 2 • ISLAND BIOGEOGRAPHY

According to island biogeography theory, which island would be expected to have the **highest** species richness?

- A Large island close to the mainland
- B Small island far from the mainland
- C Small island close to the mainland
- D Large island far from the mainland

9 UNIT 3 • SURVIVORSHIP

A species produces thousands of offspring, gives no parental care, and most young die early. Which describes it?

- A K-selected with Type I survivorship
- B K-selected with Type III survivorship
- C r-selected with Type III survivorship
- D r-selected with Type I survivorship

10 UNIT 1 • INVESTIGATION DESIGN

Students test whether leaf-litter depth affects soil moisture. Which is the best control to include?

- A Plots with no leaf litter, in the same soil and conditions
- B Plots with different amounts of sunlight
- C Plots with different soil types
- D Measuring moisture in only one plot many times

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	B - 35 years	Growth rate = $(30 - 10)/1,000 = 2\%$ per year. Rule of 70: $70 \div 2 = 35$ years. Choice A uses a 3.5% rate by mistake.
2	B - 100 kcal	About 10% passes to each level: primary consumers get 1,000 kcal, secondary consumers about 100 kcal.
3	D - 62.5 g	30 years = 3 half-lives: $500 \rightarrow 250 \rightarrow 125 \rightarrow 62.5$ g.
4	A	Nutrients cause an algal bloom; when algae die, decomposers use up dissolved oxygen, creating hypoxic water that kills fish.
5	D	CFCs release chlorine in the stratosphere, which catalytically destroys ozone, letting more UV-B through. CO_2 drives warming, not ozone loss.
6	C	Morning commute emissions of NO_x and VOCs react in sunlight, forming ground-level ozone, which peaks after midday.
7	C - \$394.20	$1.5 \text{ kW} \times 6 \text{ h} \times 365 = 3,285 \text{ kWh}$; $3,285 \times \$0.12 = \394.20 .
8	A	Large islands have more habitats and lower extinction rates; nearby islands receive more immigrants. Both increase species richness.
9	C	Many offspring, no care and high early mortality are hallmarks of r-selected species with Type III survivorship.
10	A	A control keeps everything the same except the independent variable. Plots with no leaf litter, in matching conditions, isolate its effect.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Start with energy flow, populations and the classic pollution sequences.

10 KNOW THE TARGET

How students score

In recent years, about 69% of students earned a 3 or higher on AP Environmental Science, and about 13% earned a 5. Steady preparation is what moves a score up a full point.

69%

of students earned a 3 or higher in recent years

13%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, with the charger packed
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ A **calculator** with fresh batteries (scientific or four-function is enough)
- ☐ Plan your time: about **1 min** per multiple-choice question and **22 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Review key **laws and treaties** the night before

In the free-response section: match each task verb, show setup and units for every calculation, give one specific example per point, and pair every proposed solution with a clear explanation of how it works.



ACT SAT GO • 1-ON-1 AP TUTORING

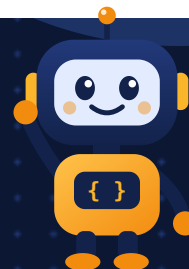
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Concept tutoring

Live 1-on-1 sessions on all nine units, with APES math drills after every lesson.

Free-response coaching

Practice all three question types and get feedback against rubric-style criteria.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

◆ Contact us

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